

# Embedding Model Comparison — Top-k Retrieval

all-MiniLM-L6-v2 vs. multilingual-e5-small vs. F2LLM-v2-80M · 50 passages, 40 queries across 10 categories · English-only, mixed general knowledge · generated 2026-07-21T09:25:53 UTC

## 1. Executive summary & recommendation

- **Recommended: all-MiniLM-L6-v2** — best in 5 of 8 comparison metrics across the 3 models on this English-only corpus.
- The models cluster near-ceiling on ranking quality here, so the deciding factors are **score separability** (how safely you can set a similarity cutoff) and operational cost (latency, model size).
- Each model sits on its own cosine scale, so each needs its **own** calibrated threshold  $\tau$  (see the calibration section) — never share a single cutoff across models.

| Metric             | all-MiniLM-L6-v2 | multilingual-e5-small | F2LLM-v2-80M | Best                  |
|--------------------|------------------|-----------------------|--------------|-----------------------|
| nDCG@10            | 0.979            | 0.975                 | 0.972        | all-MiniLM-L6-v2      |
| Recall@5           | 0.980            | 0.990                 | 0.990        | tie                   |
| MRR                | 0.985            | 0.971                 | 0.971        | all-MiniLM-L6-v2      |
| Score separability | 0.300            | 0.059                 | 0.269        | all-MiniLM-L6-v2      |
| Average Precision  | 0.847            | 0.817                 | 0.823        | all-MiniLM-L6-v2      |
| Dynamic-k F1       | 0.358            | 0.366                 | 0.364        | multilingual-e5-small |
| False-answer rate  | 0.167            | 0.000                 | 0.167        | multilingual-e5-small |
| Query latency (ms) | 1.148            | 2.475                 | 5.343        | all-MiniLM-L6-v2      |
| Metrics won        | 5                | 2                     | 0            |                       |

## 2. Setup

Each model uses its own prefix convention and L2-normalized cosine similarity. Same corpus, queries, qrels, and k grid for both. Run on CPU.

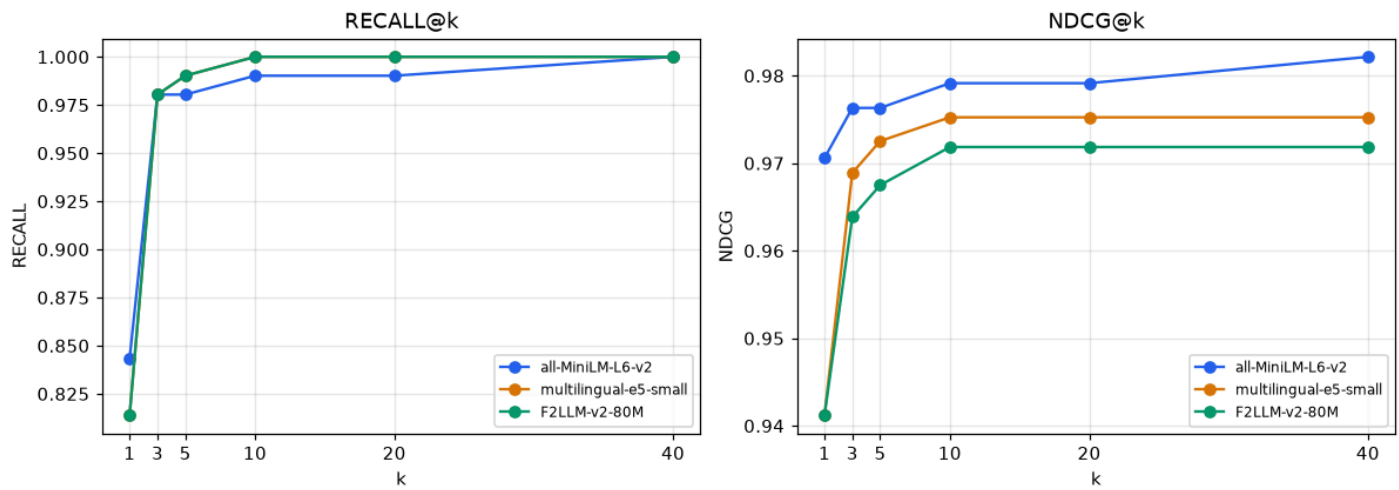
|                           | all-MiniLM-L6-v2 | multilingual-e5-small | F2LLM-v2-80M                                                                                |
|---------------------------|------------------|-----------------------|---------------------------------------------------------------------------------------------|
| Params                    | ~22M             | ~118M                 | ~80M                                                                                        |
| Embedding dim             | 384              | 384                   | 320                                                                                         |
| Query prefix              | (none)           | 'query: '             | 'Instruct: Given a question, retrieve passages that can help answer the question.\nQuery: ' |
| Passage prefix            | (none)           | 'passage: '           | (none)                                                                                      |
| Corpus encode / item (ms) | 3.36             | 8.68                  | 9.32                                                                                        |
| Query encode / item (ms)  | 1.15             | 2.48                  | 5.34                                                                                        |

**Fairness note.** multilingual-e5-small requires query: / passage: prefixes; all-MiniLM-L6-v2 uses none. Both are applied automatically. The corpus is **English-only**, so e5's multilingual strength is *not* exercised here — this is a pure English semantic-retrieval comparison.

### 3. Aggregate metrics

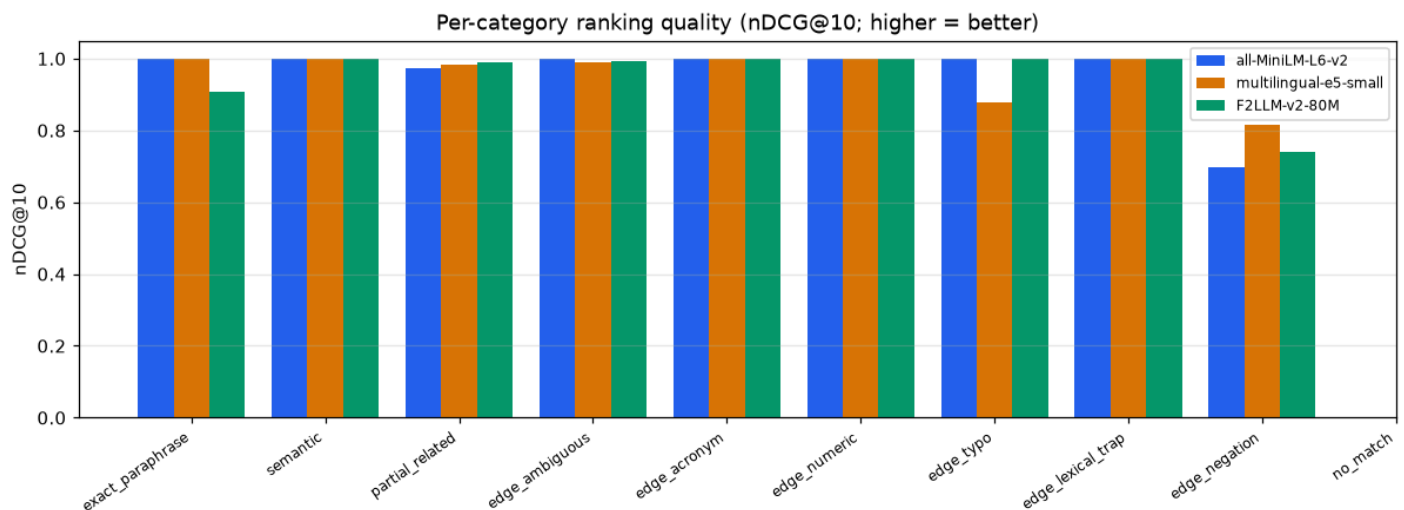
Averaged over answerable queries; best value per row highlighted. k grid: [1, 3, 5, 10, 20, 40].

| Metric       | all-MiniLM-L6-v2 | multilingual-e5-small | F2LLM-v2-80M |
|--------------|------------------|-----------------------|--------------|
| RECALL@1     | 0.843            | 0.814                 | 0.814        |
| RECALL@3     | 0.980            | 0.980                 | 0.980        |
| RECALL@5     | 0.980            | 0.990                 | 0.990        |
| RECALL@10    | 0.990            | 1.000                 | 1.000        |
| RECALL@20    | 0.990            | 1.000                 | 1.000        |
| RECALL@40    | 1.000            | 1.000                 | 1.000        |
| PRECISION@1  | 0.825            | 0.800                 | 0.800        |
| PRECISION@3  | 0.367            | 0.367                 | 0.367        |
| PRECISION@5  | 0.220            | 0.225                 | 0.225        |
| PRECISION@10 | 0.113            | 0.115                 | 0.115        |
| PRECISION@20 | 0.056            | 0.058                 | 0.058        |
| PRECISION@40 | 0.029            | 0.029                 | 0.029        |
| HIT@1        | 0.971            | 0.941                 | 0.941        |
| HIT@3        | 1.000            | 1.000                 | 1.000        |
| HIT@5        | 1.000            | 1.000                 | 1.000        |
| HIT@10       | 1.000            | 1.000                 | 1.000        |
| HIT@20       | 1.000            | 1.000                 | 1.000        |
| HIT@40       | 1.000            | 1.000                 | 1.000        |
| NDCG@1       | 0.971            | 0.941                 | 0.941        |
| NDCG@3       | 0.976            | 0.969                 | 0.964        |
| NDCG@5       | 0.976            | 0.973                 | 0.967        |
| NDCG@10      | 0.979            | 0.975                 | 0.972        |
| NDCG@20      | 0.979            | 0.975                 | 0.972        |
| NDCG@40      | 0.982            | 0.975                 | 0.972        |
| MRR          | 0.985            | 0.971                 | 0.971        |
| Separability | 0.300            | 0.059                 | 0.269        |



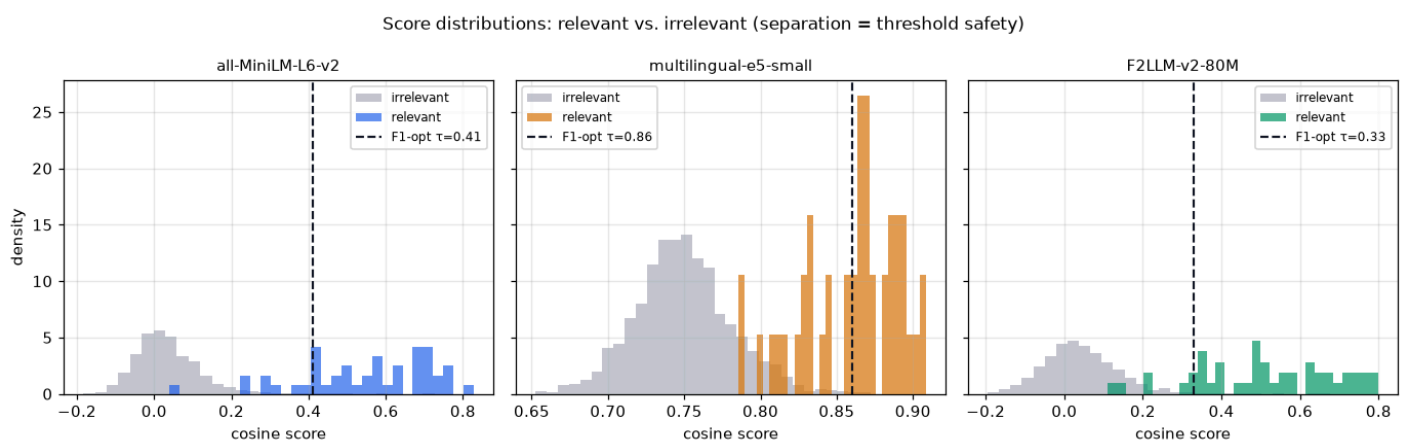
## 4. Per-category performance

Where each model wins or loses, especially on the adversarial edge cases.



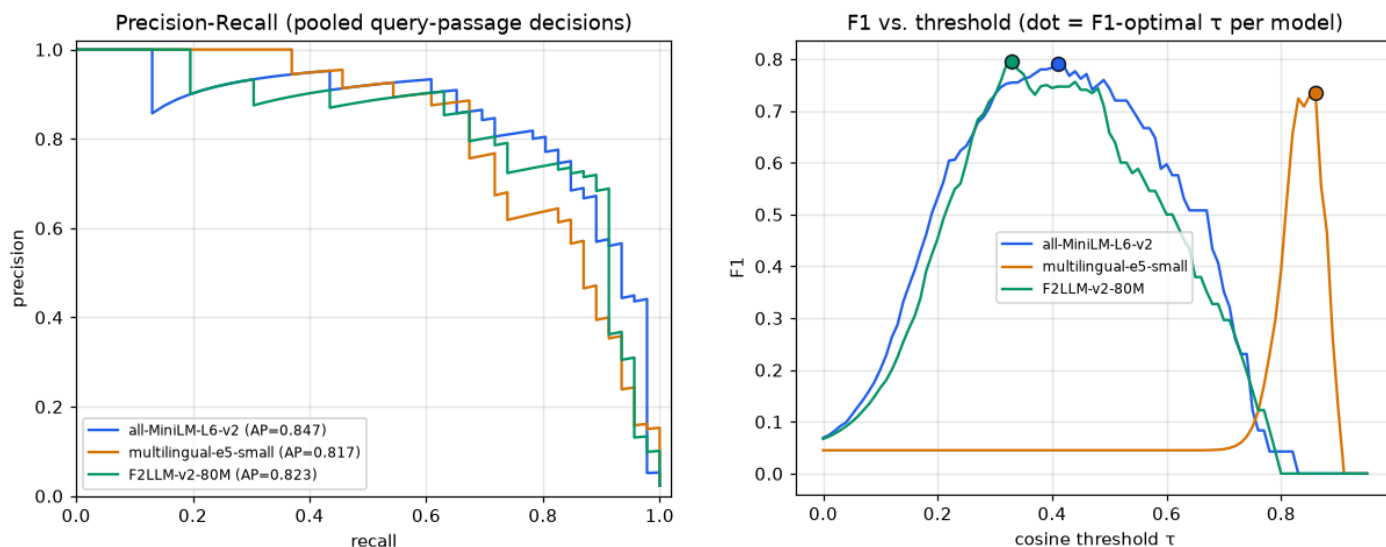
## 5. Score-distribution & separability

How well each model separates relevant from irrelevant chunks. A wide gap means a similarity cutoff is safe; overlapping distributions mean thresholding is risky. The dashed line is each model's F1-optimal  $\tau$ .



## 6. Threshold calibration & dynamic-k policy

Per-model cosine cutoffs (they differ because the models sit on different score scales) and the deployable retrieval rule: keep chunks with score  $\geq \tau$ , clamped to [5, 40] chunks.



| Operating point                | all-MiniLM-L6-v2                                                                                                      | multilingual-e5-small                                                                                                 | F2LLM-v2-80M                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| f1_optimal                     | $\tau=0.41 \cdot P=0.760 \cdot R=0.826$<br>$\cdot F1=0.792$                                                           | $\tau=0.86 \cdot P=0.879 \cdot R=0.630$<br>$\cdot F1=0.734$                                                           | $\tau=0.33 \cdot P=0.719 \cdot R=0.891$<br>$\cdot F1=0.796$                                                           |
| high_precision@0.9             | $\tau=0.49 \cdot P=0.909 \cdot R=0.652$<br>$\cdot F1=0.759$                                                           | $\tau=0.87 \cdot P=0.947 \cdot R=0.391$<br>$\cdot F1=0.554$                                                           | $\tau=0.48 \cdot P=0.906 \cdot R=0.630$<br>$\cdot F1=0.744$                                                           |
| high_recall@0.8                | $\tau=0.42 \cdot P=0.771 \cdot R=0.804$<br>$\cdot F1=0.787$                                                           | $\tau=0.83 \cdot P=0.644 \cdot R=0.826$<br>$\cdot F1=0.724$                                                           | $\tau=0.35 \cdot P=0.740 \cdot R=0.804$<br>$\cdot F1=0.771$                                                           |
| <b>Dynamic-k policy [5–40]</b> | $\tau=0.41 \cdot P=0.220 \cdot R=0.957$<br>$\cdot F1=0.358$<br>mean chunks=5.00 (range 5–5) · false-answer rate=0.167 | $\tau=0.86 \cdot P=0.225 \cdot R=0.978$<br>$\cdot F1=0.366$<br>mean chunks=5.00 (range 5–5) · false-answer rate=0.000 | $\tau=0.33 \cdot P=0.224 \cdot R=0.978$<br>$\cdot F1=0.364$<br>mean chunks=5.03 (range 5–6) · false-answer rate=0.167 |

The **dynamic-k precision** is bounded by the `k_min=5` floor: when a query has only 1–2 relevant passages, returning at least 5 chunks necessarily includes irrelevant ones. The floor is a deliberate trade to avoid starving the LLM context; recall and the false-answer rate are the more meaningful numbers for this policy.

## 7. Per-query side-by-side deep dive

Top-5 retrieved chunks per model with cosine scores. Green = a relevant (gold) passage. Ordered by category; queries where the models disagree on the #1 result are badged.

exact\_paraphrase

How do plants make their own food using sunlight?

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 2

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p01</b> 0.672<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.886<br>Photosynthesis is the process by which green plants and some other org | p38 0.548<br>Solar panels convert sunlight directly into electricity using photovol        |
| 2 | p38 0.395<br>Solar panels convert sunlight directly into electricity using photovol        | p38 0.872<br>Solar panels convert sunlight directly into electricity using photovol        | <b>p01</b> 0.536<br>Photosynthesis is the process by which green plants and some other org |
| 3 | p40 0.288<br>Honeybees are insects that pollinate flowering plants as they collect         | p34 0.835<br>Tea is an aromatic beverage made by pouring hot water over cured leave        | p04 0.302<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i        |
| 4 | p02 0.183<br>The water cycle describes how water evaporates from oceans and lakes,         | p02 0.828<br>The water cycle describes how water evaporates from oceans and lakes,         | p05 0.264<br>Gravity is the force of attraction between masses. Isaac Newton descri        |
| 5 | p39 0.172<br>Recycling is the process of collecting and reprocessing used materials        | p40 0.821<br>Honeybees are insects that pollinate flowering plants as they collect         | p37 0.253<br>Electric cars are powered by rechargeable batteries instead of petrol         |

**exact\_paraphrase** When did humans first walk on the Moon?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p09</b> 0.741<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.889<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.719<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  |
| 2 | <b>p50</b> 0.260<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p50</b> 0.781<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p32</b> 0.219<br>The World Wide Web is a system of interlinked documents and resources  |
| 3 | <b>p12</b> 0.214<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p10</b> 0.777<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p02</b> 0.180<br>The water cycle describes how water evaporates from oceans and lakes,  |
| 4 | <b>p10</b> 0.193<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p06</b> 0.773<br>According to Albert Einstein's theory of relativity, the speed of ligh | <b>p10</b> 0.173<br>Mars is the fourth planet from the Sun and is often called the Red Pla |
| 5 | <b>p05</b> 0.130<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p36</b> 0.773<br>The modern Olympic Games are an international sporting event held ever | <b>p50</b> 0.161<br>In 1903 the Wright brothers achieved the first controlled, powered, su |

exact\_paraphrase

## What is the tallest mountain on Earth?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p19</b> 0.703<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p19</b> 0.880<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p19</b> 0.754<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta |
| 2 | <b>p47</b> 0.596<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.862<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.632<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin |
| 3 | <b>p20</b> 0.293<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p20</b> 0.801<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p20</b> 0.283<br>The Sahara is the largest hot desert in the world, covering much of No |
| 4 | <b>p22</b> 0.282<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p18</b> 0.795<br>The Amazon rainforest in South America is the largest tropical rainfor | <b>p48</b> 0.260<br>Debate continues over the world's longest river: the Nile in Africa is |
| 5 | <b>p48</b> 0.271<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p22</b> 0.793<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p18</b> 0.260<br>The Amazon rainforest in South America is the largest tropical rainfor |

**exact\_paraphrase** Why was the Great Wall of China built?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p15</b> 0.744<br>The Great Wall of China was built over many centuries to defend Chines | <b>p15</b> 0.908<br>The Great Wall of China was built over many centuries to defend Chines | <b>p15</b> 0.798<br>The Great Wall of China was built over many centuries to defend Chines |
| 2 | <b>p14</b> 0.190<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p21</b> 0.794<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p42</b> 0.241<br>The Great Barrier Reef off the coast of Australia is the world's large |
| 3 | <b>p32</b> 0.150<br>The World Wide Web is a system of interlinked documents and resources  | <b>p14</b> 0.783<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p14</b> 0.163<br>The Roman Empire was one of the largest empires in ancient history, ru |
| 4 | <b>p42</b> 0.145<br>The Great Barrier Reef off the coast of Australia is the world's large | <b>p42</b> 0.776<br>The Great Barrier Reef off the coast of Australia is the world's large | <b>p32</b> 0.155<br>The World Wide Web is a system of interlinked documents and resources  |
| 5 | <b>p27</b> 0.132<br>The bank of a river is the sloping land alongside the water channel. R | <b>p08</b> 0.773<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p24</b> 0.118<br>The reticulated python is one of the world's longest snakes, a non-ven |

**semantic** What organ pushes blood around the body?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p03</b> 0.606<br>The human heart is a muscular organ that pumps blood through the circu | <b>p03</b> 0.867<br>The human heart is a muscular organ that pumps blood through the circu | <b>p03</b> 0.563<br>The human heart is a muscular organ that pumps blood through the circu |
| 2 | <b>p07</b> 0.234<br>Vaccines train the immune system to recognise and fight specific patho | <b>p07</b> 0.809<br>Vaccines train the immune system to recognise and fight specific patho | <b>p07</b> 0.222<br>Vaccines train the immune system to recognise and fight specific patho |
| 3 | <b>p25</b> 0.186<br>Java is a widely used object-oriented programming language designed to | <b>p04</b> 0.803<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | <b>p28</b> 0.216<br>A bank is a financial institution that accepts deposits from customers |
| 4 | <b>p05</b> 0.170<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p01</b> 0.772<br>Photosynthesis is the process by which green plants and some other org | <b>p11</b> 0.207<br>A black hole is a region of spacetime where gravity is so strong that  |
| 5 | <b>p21</b> 0.130<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p37</b> 0.767<br>Electric cars are powered by rechargeable batteries instead of petrol  | <b>p04</b> 0.194<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i |

**semantic** Why is the planet heating up?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p08</b> 0.494<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p08</b> 0.883<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p08</b> 0.511<br>Global warming is largely driven by greenhouse gases such as carbon di |
| 2 | <b>p10</b> 0.372<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p10</b> 0.865<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p05</b> 0.370<br>Gravity is the force of attraction between masses. Isaac Newton descri |
| 3 | <b>p20</b> 0.306<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p02</b> 0.854<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p10</b> 0.289<br>Mars is the fourth planet from the Sun and is often called the Red Pla |
| 4 | <b>p02</b> 0.291<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p38</b> 0.845<br>Solar panels convert sunlight directly into electricity using photovol | <b>p01</b> 0.257<br>Photosynthesis is the process by which green plants and some other org |
| 5 | <b>p05</b> 0.257<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p20</b> 0.845<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p38</b> 0.254<br>Solar panels convert sunlight directly into electricity using photovol |

semantic

## Which coding language was named after a comedy troupe?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p23</b> 0.446<br>Python is a popular high-level programming language known for its read | <b>p23</b> 0.834<br>Python is a popular high-level programming language known for its read | <b>p23</b> 0.386<br>Python is a popular high-level programming language known for its read |
| 2 | <b>p44</b> 0.213<br>William Shakespeare was an English playwright and poet regarded as one | <b>p25</b> 0.761<br>Java is a widely used object-oriented programming language designed to | <b>p25</b> 0.159<br>Java is a widely used object-oriented programming language designed to |
| 3 | <b>p25</b> 0.163<br>Java is a widely used object-oriented programming language designed to | <b>p44</b> 0.745<br>William Shakespeare was an English playwright and poet regarded as one | <b>p44</b> 0.147<br>William Shakespeare was an English playwright and poet regarded as one |
| 4 | <b>p50</b> 0.099<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p32</b> 0.743<br>The World Wide Web is a system of interlinked documents and resources  | <b>p36</b> 0.115<br>The modern Olympic Games are an international sporting event held ever |
| 5 | <b>p32</b> 0.094<br>The World Wide Web is a system of interlinked documents and resources  | <b>p46</b> 0.736<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p50</b> 0.113<br>In 1903 the Wright brothers achieved the first controlled, powered, su |

semantic

## How can we generate electricity from the sun?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p38</b> 0.625<br>Solar panels convert sunlight directly into electricity using photovol | <b>p38</b> 0.896<br>Solar panels convert sunlight directly into electricity using photovol | <b>p38</b> 0.641<br>Solar panels convert sunlight directly into electricity using photovol |
| 2 | <b>p01</b> 0.395<br>Photosynthesis is the process by which green plants and some other org | <b>p37</b> 0.854<br>Electric cars are powered by rechargeable batteries instead of petrol  | <b>p37</b> 0.408<br>Electric cars are powered by rechargeable batteries instead of petrol  |
| 3 | <b>p37</b> 0.313<br>Electric cars are powered by rechargeable batteries instead of petrol  | <b>p01</b> 0.849<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.371<br>Photosynthesis is the process by which green plants and some other org |
| 4 | <b>p40</b> 0.214<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p02</b> 0.828<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p08</b> 0.246<br>Global warming is largely driven by greenhouse gases such as carbon di |
| 5 | <b>p02</b> 0.192<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p05</b> 0.826<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p05</b> 0.244<br>Gravity is the force of attraction between masses. Isaac Newton descri |

**semantic** Which insects help flowers reproduce?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p40</b> 0.508<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p40</b> 0.869<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p40</b> 0.508<br>Honeybees are insects that pollinate flowering plants as they collect  |
| 2 | <b>p01</b> 0.252<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.781<br>Photosynthesis is the process by which green plants and some other org | <b>p38</b> 0.242<br>Solar panels convert sunlight directly into electricity using photovol |
| 3 | <b>p05</b> 0.168<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p18</b> 0.764<br>The Amazon rainforest in South America is the largest tropical rainfor | <b>p05</b> 0.195<br>Gravity is the force of attraction between masses. Isaac Newton descri |
| 4 | <b>p18</b> 0.155<br>The Amazon rainforest in South America is the largest tropical rainfor | <b>p24</b> 0.762<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p37</b> 0.194<br>Electric cars are powered by rechargeable batteries instead of petrol  |
| 5 | <b>p26</b> 0.117<br>Java is a densely populated island of Indonesia and home to the capita | <b>p04</b> 0.762<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | <b>p01</b> 0.181<br>Photosynthesis is the process by which green plants and some other org |

partial\_related

**What does caffeine do to the body?**

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p49</b> 0.720<br>Caffeine is a natural stimulant that acts on the central nervous syste | <b>p49</b> 0.889<br>Caffeine is a natural stimulant that acts on the central nervous syste | <b>p49</b> 0.696<br>Caffeine is a natural stimulant that acts on the central nervous syste |
| 2 | <b>p33</b> 0.532<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | <b>p33</b> 0.864<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | <b>p33</b> 0.551<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I |
| 3 | <b>p34</b> 0.399<br>Tea is an aromatic beverage made by pouring hot water over cured leave | <b>p34</b> 0.834<br>Tea is an aromatic beverage made by pouring hot water over cured leave | <b>p34</b> 0.337<br>Tea is an aromatic beverage made by pouring hot water over cured leave |
| 4 | <b>p03</b> 0.213<br>The human heart is a muscular organ that pumps blood through the circu | <b>p03</b> 0.815<br>The human heart is a muscular organ that pumps blood through the circu | <b>p03</b> 0.284<br>The human heart is a muscular organ that pumps blood through the circu |
| 5 | <b>p01</b> 0.167<br>Photosynthesis is the process by which green plants and some other org | <b>p07</b> 0.805<br>Vaccines train the immune system to recognise and fight specific patho | <b>p07</b> 0.266<br>Vaccines train the immune system to recognise and fight specific patho |

**partial\_related** Famous paintings by Italian artists

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p43</b> 0.584<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  | <b>p43</b> 0.863<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  | <b>p43</b> 0.510<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  |
| 2 | <b>p44</b> 0.149<br>William Shakespeare was an English playwright and poet regarded as one | <b>p44</b> 0.787<br>William Shakespeare was an English playwright and poet regarded as one | <b>p44</b> 0.167<br>William Shakespeare was an English playwright and poet regarded as one |
| 3 | <b>p24</b> 0.108<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p14</b> 0.763<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p49</b> 0.163<br>Caffeine is a natural stimulant that acts on the central nervous syste |
| 4 | <b>p15</b> 0.107<br>The Great Wall of China was built over many centuries to defend Chines | <b>p15</b> 0.754<br>The Great Wall of China was built over many centuries to defend Chines | <b>p23</b> 0.161<br>Python is a popular high-level programming language known for its read |
| 5 | <b>p14</b> 0.103<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p16</b> 0.751<br>The French Revolution began in 1789 and overthrew the monarchy in Fran | <b>p33</b> 0.140<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I |

**partial\_related** Tall mountains in Africa

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p47</b> 0.693<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.887<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.636<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin |
| 2 | <b>p19</b> 0.507<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p19</b> 0.842<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p48</b> 0.347<br>Debate continues over the world's longest river: the Nile in Africa is |
| 3 | <b>p20</b> 0.476<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p20</b> 0.840<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p20</b> 0.342<br>The Sahara is the largest hot desert in the world, covering much of No |
| 4 | <b>p48</b> 0.422<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p48</b> 0.835<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p19</b> 0.302<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta |
| 5 | <b>p21</b> 0.326<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p21</b> 0.814<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p21</b> 0.279<br>The Nile is a major river in northeastern Africa that flows northward  |

partial\_related

## What is the longest river in the world?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p48</b> 0.679<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p48</b> 0.888<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p48</b> 0.723<br>Debate continues over the world's longest river: the Nile in Africa is |
| 2 | <b>p27</b> 0.425<br>The bank of a river is the sloping land alongside the water channel. R | <b>p21</b> 0.816<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p21</b> 0.496<br>The Nile is a major river in northeastern Africa that flows northward  |
| 3 | <b>p21</b> 0.413<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p42</b> 0.814<br>The Great Barrier Reef off the coast of Australia is the world's large | <b>p27</b> 0.433<br>The bank of a river is the sloping land alongside the water channel. R |
| 4 | <b>p24</b> 0.327<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p22</b> 0.812<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p20</b> 0.380<br>The Sahara is the largest hot desert in the world, covering much of No |
| 5 | <b>p22</b> 0.292<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p47</b> 0.800<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p18</b> 0.349<br>The Amazon rainforest in South America is the largest tropical rainfor |

**edge\_ambiguous** Tell me about oceans and coral reefs

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p42</b> 0.572<br>The Great Barrier Reef off the coast of Australia is the world's large | <b>p42</b> 0.870<br>The Great Barrier Reef off the coast of Australia is the world's large | <b>p42</b> 0.440<br>The Great Barrier Reef off the coast of Australia is the world's large |
| 2 | <b>p22</b> 0.425<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p22</b> 0.866<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p22</b> 0.400<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc |
| 3 | <b>p41</b> 0.285<br>Sharks are a group of cartilaginous fish with a powerful sense of smel | <b>p02</b> 0.854<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p41</b> 0.219<br>Sharks are a group of cartilaginous fish with a powerful sense of smel |
| 4 | <b>p08</b> 0.271<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p08</b> 0.844<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p10</b> 0.219<br>Mars is the fourth planet from the Sun and is often called the Red Pla |
| 5 | <b>p18</b> 0.265<br>The Amazon rainforest in South America is the largest tropical rainfor | <b>p41</b> 0.831<br>Sharks are a group of cartilaginous fish with a powerful sense of smel | <b>p02</b> 0.216<br>The water cycle describes how water evaporates from oceans and lakes,  |

edge\_ambiguous

## Popular hot caffeinated beverages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p34</b> 0.584<br>Tea is an aromatic beverage made by pouring hot water over cured leave | <b>p33</b> 0.868<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | <b>p33</b> 0.617<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I |
| 2 | <b>p33</b> 0.551<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | <b>p34</b> 0.830<br>Tea is an aromatic beverage made by pouring hot water over cured leave | <b>p49</b> 0.480<br>Caffeine is a natural stimulant that acts on the central nervous syste |
| 3 | <b>p49</b> 0.462<br>Caffeine is a natural stimulant that acts on the central nervous syste | <b>p49</b> 0.827<br>Caffeine is a natural stimulant that acts on the central nervous syste | <b>p34</b> 0.367<br>Tea is an aromatic beverage made by pouring hot water over cured leave |
| 4 | <b>p47</b> 0.156<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p20</b> 0.766<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p43</b> 0.188<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  |
| 5 | <b>p20</b> 0.127<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p50</b> 0.762<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p18</b> 0.188<br>The Amazon rainforest in South America is the largest tropical rainfor |

**edge\_ambiguous** Facts about famous mountains

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p47</b> 0.426<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.847<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p47</b> 0.354<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin |
| 2 | <b>p19</b> 0.423<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p19</b> 0.845<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p19</b> 0.338<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta |
| 3 | <b>p20</b> 0.294<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p15</b> 0.810<br>The Great Wall of China was built over many centuries to defend Chines | <b>p43</b> 0.314<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  |
| 4 | <b>p15</b> 0.222<br>The Great Wall of China was built over many centuries to defend Chines | <b>p20</b> 0.809<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p14</b> 0.295<br>The Roman Empire was one of the largest empires in ancient history, ru |
| 5 | <b>p48</b> 0.218<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p43</b> 0.803<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  | <b>p23</b> 0.270<br>Python is a popular high-level programming language known for its read |

edge\_ambiguous

## Widely used programming languages

disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p23</b> 0.503<br>Python is a popular high-level programming language known for its read | <b>p25</b> 0.870<br>Java is a widely used object-oriented programming language designed to | <b>p23</b> 0.465<br>Python is a popular high-level programming language known for its read |
| 2 | <b>p25</b> 0.468<br>Java is a widely used object-oriented programming language designed to | <b>p23</b> 0.866<br>Python is a popular high-level programming language known for its read | <b>p25</b> 0.402<br>Java is a widely used object-oriented programming language designed to |
| 3 | <b>p32</b> 0.303<br>The World Wide Web is a system of interlinked documents and resources  | <b>p31</b> 0.801<br>Machine learning is a branch of artificial intelligence in which compu | <b>p32</b> 0.321<br>The World Wide Web is a system of interlinked documents and resources  |
| 4 | <b>p14</b> 0.174<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p32</b> 0.794<br>The World Wide Web is a system of interlinked documents and resources  | <b>p44</b> 0.315<br>William Shakespeare was an English playwright and poet regarded as one |
| 5 | <b>p29</b> 0.169<br>Cloud computing delivers computing services such as storage and proces | <b>p44</b> 0.789<br>William Shakespeare was an English playwright and poet regarded as one | <b>p14</b> 0.202<br>The Roman Empire was one of the largest empires in ancient history, ru |

edge\_acronym

## What is NASA?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p12</b> 0.775<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p12</b> 0.896<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p12</b> 0.743<br>NASA, the National Aeronautics and Space Administration, is the United |
| 2 | <b>p10</b> 0.362<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p10</b> 0.810<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p10</b> 0.315<br>Mars is the fourth planet from the Sun and is often called the Red Pla |
| 3 | <b>p09</b> 0.261<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p30</b> 0.805<br>A cloud is a visible mass of water droplets or ice crystals suspended  | <b>p05</b> 0.304<br>Gravity is the force of attraction between masses. Isaac Newton descri |
| 4 | <b>p05</b> 0.229<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p32</b> 0.804<br>The World Wide Web is a system of interlinked documents and resources  | <b>p21</b> 0.251<br>The Nile is a major river in northeastern Africa that flows northward  |
| 5 | <b>p40</b> 0.228<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p05</b> 0.802<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p32</b> 0.239<br>The World Wide Web is a system of interlinked documents and resources  |

**edge\_acronym** The American government space agency

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p12</b> 0.749<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p12</b> 0.896<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p12</b> 0.736<br>NASA, the National Aeronautics and Space Administration, is the United |
| 2 | <b>p09</b> 0.203<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p05</b> 0.797<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p18</b> 0.165<br>The Amazon rainforest in South America is the largest tropical rainfor |
| 3 | <b>p10</b> 0.180<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p50</b> 0.788<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p05</b> 0.158<br>Gravity is the force of attraction between masses. Isaac Newton descri |
| 4 | <b>p32</b> 0.178<br>The World Wide Web is a system of interlinked documents and resources  | <b>p11</b> 0.778<br>A black hole is a region of spacetime where gravity is so strong that  | <b>p32</b> 0.148<br>The World Wide Web is a system of interlinked documents and resources  |
| 5 | <b>p29</b> 0.172<br>Cloud computing delivers computing services such as storage and proces | <b>p10</b> 0.775<br>Mars is the fourth planet from the Sun and is often called the Red Pla | <b>p20</b> 0.126<br>The Sahara is the largest hot desert in the world, covering much of No |

edge\_acronym

## What does DNA do in living things?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p04</b> 0.673<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | <b>p04</b> 0.884<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | <b>p04</b> 0.663<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i |
| 2 | <b>p03</b> 0.289<br>The human heart is a muscular organ that pumps blood through the circu | <b>p01</b> 0.822<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.318<br>Photosynthesis is the process by which green plants and some other org |
| 3 | <b>p01</b> 0.285<br>Photosynthesis is the process by which green plants and some other org | <b>p07</b> 0.807<br>Vaccines train the immune system to recognise and fight specific patho | <b>p28</b> 0.263<br>A bank is a financial institution that accepts deposits from customers |
| 4 | <b>p07</b> 0.203<br>Vaccines train the immune system to recognise and fight specific patho | <b>p40</b> 0.801<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p39</b> 0.250<br>Recycling is the process of collecting and reprocessing used materials |
| 5 | <b>p41</b> 0.192<br>Sharks are a group of cartilaginous fish with a powerful sense of smel | <b>p38</b> 0.799<br>Solar panels convert sunlight directly into electricity using photovol | <b>p29</b> 0.248<br>Cloud computing delivers computing services such as storage and proces |

edge\_numeric

**What major event happened in 1969?**

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p09</b> 0.290<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.788<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.355<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  |
| 2 | <b>p50</b> 0.260<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p36</b> 0.755<br>The modern Olympic Games are an international sporting event held ever | <b>p13</b> 0.212<br>World War II ended in 1945 after six years of global conflict. Victory |
| 3 | <b>p16</b> 0.250<br>The French Revolution began in 1789 and overthrew the monarchy in Fran | <b>p50</b> 0.750<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p16</b> 0.151<br>The French Revolution began in 1789 and overthrew the monarchy in Fran |
| 4 | <b>p13</b> 0.198<br>World War II ended in 1945 after six years of global conflict. Victory | <b>p17</b> 0.745<br>The Industrial Revolution was a period of rapid technological and econ | <b>p17</b> 0.131<br>The Industrial Revolution was a period of rapid technological and econ |
| 5 | <b>p36</b> 0.191<br>The modern Olympic Games are an international sporting event held ever | <b>p21</b> 0.735<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p50</b> 0.105<br>In 1903 the Wright brothers achieved the first controlled, powered, su |

edge\_numeric

## What is the value of pi to five decimal places?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p45</b> 0.700<br>Pi is the ratio of a circle's circumference to its diameter, approxima | <b>p45</b> 0.856<br>Pi is the ratio of a circle's circumference to its diameter, approxima | <b>p45</b> 0.591<br>Pi is the ratio of a circle's circumference to its diameter, approxima |
| 2 | <b>p46</b> 0.100<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p23</b> 0.800<br>Python is a popular high-level programming language known for its read | <b>p22</b> 0.172<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc |
| 3 | <b>p22</b> 0.094<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p46</b> 0.779<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p46</b> 0.142<br>The Pythagorean theorem states that in a right-angled triangle the squ |
| 4 | <b>p43</b> 0.086<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  | <b>p02</b> 0.773<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p05</b> 0.098<br>Gravity is the force of attraction between masses. Isaac Newton descri |
| 5 | <b>p23</b> 0.082<br>Python is a popular high-level programming language known for its read | <b>p06</b> 0.769<br>According to Albert Einstein's theory of relativity, the speed of ligh | <b>p26</b> 0.097<br>Java is a densely populated island of Indonesia and home to the capita |

**edge\_numeric** Which world war ended in 1945?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p13</b> 0.759<br>World War II ended in 1945 after six years of global conflict. Victory | <b>p13</b> 0.902<br>World War II ended in 1945 after six years of global conflict. Victory | <b>p13</b> 0.789<br>World War II ended in 1945 after six years of global conflict. Victory |
| 2 | <b>p32</b> 0.183<br>The World Wide Web is a system of interlinked documents and resources  | <b>p32</b> 0.776<br>The World Wide Web is a system of interlinked documents and resources  | <b>p32</b> 0.213<br>The World Wide Web is a system of interlinked documents and resources  |
| 3 | <b>p22</b> 0.150<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p15</b> 0.762<br>The Great Wall of China was built over many centuries to defend Chines | <b>p36</b> 0.185<br>The modern Olympic Games are an international sporting event held ever |
| 4 | <b>p36</b> 0.117<br>The modern Olympic Games are an international sporting event held ever | <b>p36</b> 0.757<br>The modern Olympic Games are an international sporting event held ever | <b>p14</b> 0.173<br>The Roman Empire was one of the largest empires in ancient history, ru |
| 5 | <b>p02</b> 0.116<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p16</b> 0.743<br>The French Revolution began in 1789 and overthrew the monarchy in Fran | <b>p39</b> 0.171<br>Recycling is the process of collecting and reprocessing used materials |

edge\_numeric

## In what year was the first powered airplane flight?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p50</b> 0.680<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p50</b> 0.862<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p50</b> 0.627<br>In 1903 the Wright brothers achieved the first controlled, powered, su |
| 2 | <b>p09</b> 0.350<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.765<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p09</b> 0.394<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  |
| 3 | <b>p12</b> 0.172<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p17</b> 0.755<br>The Industrial Revolution was a period of rapid technological and econ | <b>p16</b> 0.220<br>The French Revolution began in 1789 and overthrew the monarchy in Fran |
| 4 | <b>p17</b> 0.149<br>The Industrial Revolution was a period of rapid technological and econ | <b>p37</b> 0.746<br>Electric cars are powered by rechargeable batteries instead of petrol  | <b>p17</b> 0.191<br>The Industrial Revolution was a period of rapid technological and econ |
| 5 | <b>p13</b> 0.133<br>World War II ended in 1945 after six years of global conflict. Victory | <b>p32</b> 0.737<br>The World Wide Web is a system of interlinked documents and resources  | <b>p32</b> 0.158<br>The World Wide Web is a system of interlinked documents and resources  |

**edge\_type** how does fotosynthesis wrok in plants

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p01</b> 0.583<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.869<br>Photosynthesis is the process by which green plants and some other org | <b>p01</b> 0.487<br>Photosynthesis is the process by which green plants and some other org |
| 2 | <b>p38</b> 0.264<br>Solar panels convert sunlight directly into electricity using photovol | <b>p02</b> 0.809<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p10</b> 0.241<br>Mars is the fourth planet from the Sun and is often called the Red Pla |
| 3 | <b>p40</b> 0.194<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p38</b> 0.807<br>Solar panels convert sunlight directly into electricity using photovol | <b>p02</b> 0.205<br>The water cycle describes how water evaporates from oceans and lakes,  |
| 4 | <b>p04</b> 0.167<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | <b>p46</b> 0.792<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p38</b> 0.183<br>Solar panels convert sunlight directly into electricity using photovol |
| 5 | <b>p39</b> 0.141<br>Recycling is the process of collecting and reprocessing used materials | <b>p27</b> 0.788<br>The bank of a river is the sloping land alongside the water channel. R | <b>p04</b> 0.181<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i |

edge\_typo the tallest montain in the wrld disagreement

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 2 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p19</b> 0.426<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p47</b> 0.807<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p19</b> 0.497<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta |
| 2 | <b>p47</b> 0.422<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | <b>p19</b> 0.799<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | <b>p47</b> 0.467<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin |
| 3 | <b>p20</b> 0.231<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p50</b> 0.754<br>In 1903 the Wright brothers achieved the first controlled, powered, su | <b>p20</b> 0.218<br>The Sahara is the largest hot desert in the world, covering much of No |
| 4 | <b>p14</b> 0.208<br>The Roman Empire was one of the largest empires in ancient history, ru | <b>p48</b> 0.752<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p43</b> 0.170<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  |
| 5 | <b>p22</b> 0.186<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc | <b>p09</b> 0.748<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | <b>p50</b> 0.167<br>In 1903 the Wright brothers achieved the first controlled, powered, su |

**edge\_type** explain the pythagorean theorem

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p46</b> 0.717<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p46</b> 0.894<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p46</b> 0.675<br>The Pythagorean theorem states that in a right-angled triangle the squ |
| 2 | <b>p45</b> 0.293<br>Pi is the ratio of a circle's circumference to its diameter, approxima | <b>p03</b> 0.816<br>The human heart is a muscular organ that pumps blood through the circu | <b>p23</b> 0.264<br>Python is a popular high-level programming language known for its read |
| 3 | <b>p05</b> 0.181<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p45</b> 0.796<br>Pi is the ratio of a circle's circumference to its diameter, approxima | <b>p24</b> 0.228<br>The reticulated python is one of the world's longest snakes, a non-ven |
| 4 | <b>p03</b> 0.157<br>The human heart is a muscular organ that pumps blood through the circu | <b>p07</b> 0.794<br>Vaccines train the immune system to recognise and fight specific patho | <b>p11</b> 0.197<br>A black hole is a region of spacetime where gravity is so strong that  |
| 5 | <b>p06</b> 0.134<br>According to Albert Einstein's theory of relativity, the speed of ligh | <b>p24</b> 0.793<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p38</b> 0.150<br>Solar panels convert sunlight directly into electricity using photovol |

edge\_lexical\_trap

## Using the Python programming language for data science

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p23</b> 0.628<br>Python is a popular high-level programming language known for its read | <b>p23</b> 0.875<br>Python is a popular high-level programming language known for its read | <b>p23</b> 0.582<br>Python is a popular high-level programming language known for its read |
| 2 | <b>p25</b> 0.227<br>Java is a widely used object-oriented programming language designed to | <b>p31</b> 0.818<br>Machine learning is a branch of artificial intelligence in which compu | <b>p24</b> 0.187<br>The reticulated python is one of the world's longest snakes, a non-ven |
| 3 | <b>p12</b> 0.188<br>NASA, the National Aeronautics and Space Administration, is the United | <b>p25</b> 0.811<br>Java is a widely used object-oriented programming language designed to | <b>p31</b> 0.179<br>Machine learning is a branch of artificial intelligence in which compu |
| 4 | <b>p31</b> 0.185<br>Machine learning is a branch of artificial intelligence in which compu | <b>p45</b> 0.801<br>Pi is the ratio of a circle's circumference to its diameter, approxima | <b>p44</b> 0.172<br>William Shakespeare was an English playwright and poet regarded as one |
| 5 | <b>p24</b> 0.181<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p02</b> 0.795<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p04</b> 0.172<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i |

**edge\_lexical\_trap** The grassy bank at the edge of a river

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p27</b> 0.720<br>The bank of a river is the sloping land alongside the water channel. R | <b>p27</b> 0.873<br>The bank of a river is the sloping land alongside the water channel. R | <b>p27</b> 0.647<br>The bank of a river is the sloping land alongside the water channel. R |
| 2 | <b>p21</b> 0.412<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p21</b> 0.819<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p28</b> 0.259<br>A bank is a financial institution that accepts deposits from customers |
| 3 | <b>p28</b> 0.404<br>A bank is a financial institution that accepts deposits from customers | <b>p28</b> 0.808<br>A bank is a financial institution that accepts deposits from customers | <b>p21</b> 0.233<br>The Nile is a major river in northeastern Africa that flows northward  |
| 4 | <b>p48</b> 0.357<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p48</b> 0.778<br>Debate continues over the world's longest river: the Nile in Africa is | <b>p48</b> 0.185<br>Debate continues over the world's longest river: the Nile in Africa is |
| 5 | <b>p46</b> 0.203<br>The Pythagorean theorem states that in a right-angled triangle the squ | <b>p02</b> 0.773<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p02</b> 0.130<br>The water cycle describes how water evaporates from oceans and lakes,  |

**edge\_lexical\_trap** Where can I deposit money and open an account?

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p28</b> 0.543<br>A bank is a financial institution that accepts deposits from customers | <b>p28</b> 0.864<br>A bank is a financial institution that accepts deposits from customers | <b>p28</b> 0.347<br>A bank is a financial institution that accepts deposits from customers |
| 2 | <b>p27</b> 0.210<br>The bank of a river is the sloping land alongside the water channel. R | <b>p27</b> 0.813<br>The bank of a river is the sloping land alongside the water channel. R | <b>p30</b> 0.205<br>A cloud is a visible mass of water droplets or ice crystals suspended  |
| 3 | <b>p11</b> 0.207<br>A black hole is a region of spacetime where gravity is so strong that  | <b>p21</b> 0.787<br>The Nile is a major river in northeastern Africa that flows northward  | <b>p27</b> 0.194<br>The bank of a river is the sloping land alongside the water channel. R |
| 4 | <b>p05</b> 0.079<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p03</b> 0.785<br>The human heart is a muscular organ that pumps blood through the circu | <b>p02</b> 0.177<br>The water cycle describes how water evaporates from oceans and lakes,  |
| 5 | <b>p43</b> 0.075<br>The Mona Lisa is a famous portrait painted by the Italian Renaissance  | <b>p29</b> 0.782<br>Cloud computing delivers computing services such as storage and proces | <b>p03</b> 0.173<br>The human heart is a muscular organ that pumps blood through the circu |

edge\_lexical\_trap

The island of Java in Indonesia

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                                      | multilingual-e5-small                                                                                 | F2LLM-v2-80M                                                                                          |
|---|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1 | <div>p260.828</div> <div>Java is a densely populated island of Indonesia and home to the capita</div> | <div>p260.907</div> <div>Java is a densely populated island of Indonesia and home to the capita</div> | <div>p260.773</div> <div>Java is a densely populated island of Indonesia and home to the capita</div> |
| 2 | <div>p250.486</div> <div>Java is a widely used object-oriented programming language designed to</div> | <div>p250.821</div> <div>Java is a widely used object-oriented programming language designed to</div> | <div>p250.408</div> <div>Java is a widely used object-oriented programming language designed to</div> |
| 3 | <div>p220.204</div> <div>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc</div> | <div>p220.795</div> <div>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc</div> | <div>p420.142</div> <div>The Great Barrier Reef off the coast of Australia is the world's large</div> |
| 4 | <div>p470.183</div> <div>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin</div> | <div>p420.780</div> <div>The Great Barrier Reef off the coast of Australia is the world's large</div> | <div>p180.138</div> <div>The Amazon rainforest in South America is the largest tropical rainfor</div> |
| 5 | <div>p130.172</div> <div>World War II ended in 1945 after six years of global conflict. Victory</div> | <div>p130.780</div> <div>World War II ended in 1945 after six years of global conflict. Victory</div> | <div>p130.109</div> <div>World War II ended in 1945 after six years of global conflict. Victory</div> |

**edge\_lexical\_trap** Dark clouds gathering in the sky before rain

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p30</b> 0.639<br>A cloud is a visible mass of water droplets or ice crystals suspended  | <b>p30</b> 0.857<br>A cloud is a visible mass of water droplets or ice crystals suspended  | <b>p30</b> 0.499<br>A cloud is a visible mass of water droplets or ice crystals suspended  |
| 2 | <b>p02</b> 0.343<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p02</b> 0.796<br>The water cycle describes how water evaporates from oceans and lakes,  | <b>p29</b> 0.288<br>Cloud computing delivers computing services such as storage and proces |
| 3 | <b>p08</b> 0.183<br>Global warming is largely driven by greenhouse gases such as carbon di | <b>p11</b> 0.787<br>A black hole is a region of spacetime where gravity is so strong that  | <b>p02</b> 0.205<br>The water cycle describes how water evaporates from oceans and lakes,  |
| 4 | <b>p29</b> 0.179<br>Cloud computing delivers computing services such as storage and proces | <b>p20</b> 0.778<br>The Sahara is the largest hot desert in the world, covering much of No | <b>p18</b> 0.199<br>The Amazon rainforest in South America is the largest tropical rainfor |
| 5 | <b>p11</b> 0.176<br>A black hole is a region of spacetime where gravity is so strong that  | <b>p05</b> 0.770<br>Gravity is the force of attraction between masses. Isaac Newton descri | <b>p24</b> 0.168<br>The reticulated python is one of the world's longest snakes, a non-ven |

edge\_negation

**A programming language that is not Python**

all-MiniLM-L6-v2 gold rank: 2 · multilingual-e5-small gold rank: 2 · F2LLM-v2-80M gold rank: 2

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p23 0.726<br>Python is a popular high-level programming language known for its read | p23 0.869<br>Python is a popular high-level programming language known for its read | p23 0.676<br>Python is a popular high-level programming language known for its read |
| 2 | p25 0.355<br>Java is a widely used object-oriented programming language designed to | p25 0.823<br>Java is a widely used object-oriented programming language designed to | p25 0.316<br>Java is a widely used object-oriented programming language designed to |
| 3 | p24 0.296<br>The reticulated python is one of the world's longest snakes, a non-ven | p31 0.810<br>Machine learning is a branch of artificial intelligence in which compu | p44 0.314<br>William Shakespeare was an English playwright and poet regarded as one |
| 4 | p32 0.162<br>The World Wide Web is a system of interlinked documents and resources  | p45 0.797<br>Pi is the ratio of a circle's circumference to its diameter, approxima | p24 0.202<br>The reticulated python is one of the world's longest snakes, a non-ven |
| 5 | p34 0.153<br>Tea is an aromatic beverage made by pouring hot water over cured leave | p46 0.789<br>The Pythagorean theorem states that in a right-angled triangle the squ | p31 0.201<br>Machine learning is a branch of artificial intelligence in which compu |

**edge\_negation** Common animals that are not mammals

all-MiniLM-L6-v2 gold rank: 1 · multilingual-e5-small gold rank: 1 · F2LLM-v2-80M gold rank: 1

| # | all-MiniLM-L6-v2                                                                           | multilingual-e5-small                                                                      | F2LLM-v2-80M                                                                               |
|---|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>p41</b> 0.327<br>Sharks are a group of cartilaginous fish with a powerful sense of smel | <b>p41</b> 0.841<br>Sharks are a group of cartilaginous fish with a powerful sense of smel | <b>p41</b> 0.202<br>Sharks are a group of cartilaginous fish with a powerful sense of smel |
| 2 | <b>p24</b> 0.227<br>The reticulated python is one of the world's longest snakes, a non-ven | <b>p24</b> 0.812<br>The reticulated python is one of the world's longest snakes, a non-ven | p30 0.136<br>A cloud is a visible mass of water droplets or ice crystals suspended         |
| 3 | p03 0.142<br>The human heart is a muscular organ that pumps blood through the circu        | <b>p40</b> 0.808<br>Honeybees are insects that pollinate flowering plants as they collect  | <b>p40</b> 0.133<br>Honeybees are insects that pollinate flowering plants as they collect  |
| 4 | p18 0.134<br>The Amazon rainforest in South America is the largest tropical rainfor        | p03 0.807<br>The human heart is a muscular organ that pumps blood through the circu        | p42 0.132<br>The Great Barrier Reef off the coast of Australia is the world's large        |
| 5 | p09 0.129<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the         | p37 0.804<br>Electric cars are powered by rechargeable batteries instead of petrol         | p31 0.131<br>Machine learning is a branch of artificial intelligence in which compu        |

edge\_negation

Hot drinks that contain no caffeine

disagreement

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p34 0.450<br>Tea is an aromatic beverage made by pouring hot water over cured leave | p33 0.855<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | p33 0.478<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I |
| 2 | p49 0.436<br>Caffeine is a natural stimulant that acts on the central nervous syste | p49 0.848<br>Caffeine is a natural stimulant that acts on the central nervous syste | p49 0.382<br>Caffeine is a natural stimulant that acts on the central nervous syste |
| 3 | p33 0.428<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | p34 0.841<br>Tea is an aromatic beverage made by pouring hot water over cured leave | p34 0.302<br>Tea is an aromatic beverage made by pouring hot water over cured leave |
| 4 | p20 0.118<br>The Sahara is the largest hot desert in the world, covering much of No | p37 0.776<br>Electric cars are powered by rechargeable batteries instead of petrol  | p37 0.142<br>Electric cars are powered by rechargeable batteries instead of petrol  |
| 5 | p24 0.111<br>The reticulated python is one of the world's longest snakes, a non-ven | p20 0.773<br>The Sahara is the largest hot desert in the world, covering much of No | p20 0.111<br>The Sahara is the largest hot desert in the world, covering much of No |

no\_match

## Best recipe for chocolate chip cookies

disagreement

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p50 0.133<br>In 1903 the Wright brothers achieved the first controlled, powered, su | p33 0.767<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | p22 0.099<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc |
| 2 | p09 0.114<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  | p01 0.766<br>Photosynthesis is the process by which green plants and some other org | p42 0.070<br>The Great Barrier Reef off the coast of Australia is the world's large |
| 3 | p40 0.089<br>Honeybees are insects that pollinate flowering plants as they collect  | p34 0.766<br>Tea is an aromatic beverage made by pouring hot water over cured leave | p09 0.067<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  |
| 4 | p35 0.062<br>Chess is a two-player strategy board game played on an eight-by-eight  | p23 0.761<br>Python is a popular high-level programming language known for its read | p35 0.066<br>Chess is a two-player strategy board game played on an eight-by-eight  |
| 5 | p33 0.056<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I | p46 0.758<br>The Pythagorean theorem states that in a right-angled triangle the squ | p33 0.052<br>Coffee is a popular brewed drink prepared from roasted coffee beans. I |

no\_match

How do I change a flat car tire? disagreement

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p05 0.108<br>Gravity is the force of attraction between masses. Isaac Newton descri | p37 0.785<br>Electric cars are powered by rechargeable batteries instead of petrol  | p16 0.147<br>The French Revolution began in 1789 and overthrew the monarchy in Fran |
| 2 | p16 0.099<br>The French Revolution began in 1789 and overthrew the monarchy in Fran | p17 0.775<br>The Industrial Revolution was a period of rapid technological and econ | p37 0.099<br>Electric cars are powered by rechargeable batteries instead of petrol  |
| 3 | p04 0.068<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i | p27 0.774<br>The bank of a river is the sloping land alongside the water channel. R | p08 0.095<br>Global warming is largely driven by greenhouse gases such as carbon di |
| 4 | p08 0.060<br>Global warming is largely driven by greenhouse gases such as carbon di | p02 0.771<br>The water cycle describes how water evaporates from oceans and lakes,  | p50 0.094<br>In 1903 the Wright brothers achieved the first controlled, powered, su |
| 5 | p10 0.056<br>Mars is the fourth planet from the Sun and is often called the Red Pla | p03 0.769<br>The human heart is a muscular organ that pumps blood through the circu | p02 0.081<br>The water cycle describes how water evaporates from oceans and lakes,  |

no\_match

What is the current stock price of Apple?

disagreement

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p25 0.132<br>Java is a widely used object-oriented programming language designed to | p25 0.741<br>Java is a widely used object-oriented programming language designed to | p16 0.139<br>The French Revolution began in 1789 and overthrew the monarchy in Fran |
| 2 | p29 0.089<br>Cloud computing delivers computing services such as storage and proces | p38 0.738<br>Solar panels convert sunlight directly into electricity using photovol | p45 0.135<br>Pi is the ratio of a circle's circumference to its diameter, approxima |
| 3 | p47 0.085<br>Mount Kilimanjaro in Tanzania is the highest mountain in Africa, risin | p29 0.737<br>Cloud computing delivers computing services such as storage and proces | p50 0.126<br>In 1903 the Wright brothers achieved the first controlled, powered, su |
| 4 | p06 0.084<br>According to Albert Einstein's theory of relativity, the speed of ligh | p17 0.736<br>The Industrial Revolution was a period of rapid technological and econ | p09 0.113<br>On 20 July 1969, the Apollo 11 mission landed the first humans on the  |
| 5 | p19 0.061<br>Mount Everest, on the border of Nepal and Tibet, is the highest mounta | p37 0.735<br>Electric cars are powered by rechargeable batteries instead of petrol  | p22 0.107<br>The Pacific Ocean is the largest and deepest of Earth's oceans, stretc |

no\_match

## Explain the rules of cricket batting

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p35 0.230<br>Chess is a two-player strategy board game played on an eight-by-eight  | p35 0.814<br>Chess is a two-player strategy board game played on an eight-by-eight  | p35 0.210<br>Chess is a two-player strategy board game played on an eight-by-eight  |
| 2 | p07 0.215<br>Vaccines train the immune system to recognise and fight specific patho | p03 0.810<br>The human heart is a muscular organ that pumps blood through the circu | p07 0.206<br>Vaccines train the immune system to recognise and fight specific patho |
| 3 | p40 0.201<br>Honeybees are insects that pollinate flowering plants as they collect  | p07 0.806<br>Vaccines train the immune system to recognise and fight specific patho | p31 0.187<br>Machine learning is a branch of artificial intelligence in which compu |
| 4 | p34 0.120<br>Tea is an aromatic beverage made by pouring hot water over cured leave | p46 0.791<br>The Pythagorean theorem states that in a right-angled triangle the squ | p28 0.158<br>A bank is a financial institution that accepts deposits from customers |
| 5 | p05 0.111<br>Gravity is the force of attraction between masses. Isaac Newton descri | p36 0.790<br>The modern Olympic Games are an international sporting event held ever | p50 0.146<br>In 1903 the Wright brothers achieved the first controlled, powered, su |

**no\_match** How do I house-train a puppy? **disagreement**

no relevant passage (unanswerable)

| # | all-MiniLM-L6-v2                                                                    | multilingual-e5-small                                                               | F2LLM-v2-80M                                                                        |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | p31 0.135<br>Machine learning is a branch of artificial intelligence in which compu | p07 0.795<br>Vaccines train the immune system to recognise and fight specific patho | p03 0.174<br>The human heart is a muscular organ that pumps blood through the circu |
| 2 | p50 0.109<br>In 1903 the Wright brothers achieved the first controlled, powered, su | p37 0.781<br>Electric cars are powered by rechargeable batteries instead of petrol  | p07 0.174<br>Vaccines train the immune system to recognise and fight specific patho |
| 3 | p07 0.104<br>Vaccines train the immune system to recognise and fight specific patho | p24 0.779<br>The reticulated python is one of the world's longest snakes, a non-ven | p04 0.158<br>DNA, or deoxyribonucleic acid, carries the genetic instructions used i |
| 4 | p37 0.088<br>Electric cars are powered by rechargeable batteries instead of petrol  | p03 0.773<br>The human heart is a muscular organ that pumps blood through the circu | p31 0.096<br>Machine learning is a branch of artificial intelligence in which compu |
| 5 | p03 0.048<br>The human heart is a muscular organ that pumps blood through the circu | p46 0.772<br>The Pythagorean theorem states that in a right-angled triangle the squ | p05 0.096<br>Gravity is the force of attraction between masses. Isaac Newton descri |

## 8. Caveats

- **Synthetic data.** A hand-authored 50-passage corpus is small; absolute metrics are optimistic and category-level numbers rest on a few queries each. Use the *relative* comparison, not absolute scores.
- **CPU timing.** Latencies are CPU-only and indicative, not production throughput.
- **English-only.** e5's multilingual capability is untested here; if your real workload is multilingual, re-run with a multilingual corpus before deciding.
- **Swap in real data.** Replace `data/dataset.json` with your own passages/queries/qrels and re-run `evaluate.py` + `report.py` unchanged.

Generated by the embedding comparison harness · raw data in `results.json` / `aggregate.csv` / `per_query.csv`